

AI Servo 數據分析、閉環回饋與三菱伺服參數優化白皮書

Semiconductor Automation Servo AI - v4.2 Analysis Feedback Edition

本文件整合 HMI、PLC、Servo、Motor、Encoder、EtherCAT/CC-Link、Safety、Vision AOI、Robot 與 I/O Tag，建立資料分析後的參數回饋與性能提升流程。

HMI / PLC / Servo / Motor / AI Servo closed-loop architecture

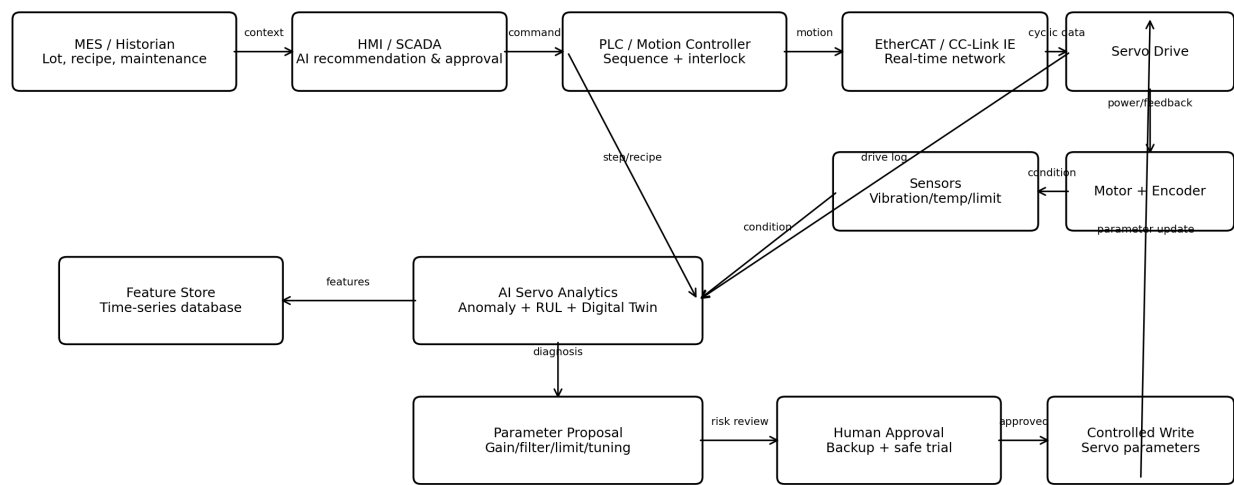


圖 1：HMI / PLC / Servo / Motor / AI Servo 通訊與閉環架構

AI Servo data analysis -> parameter feedback loop

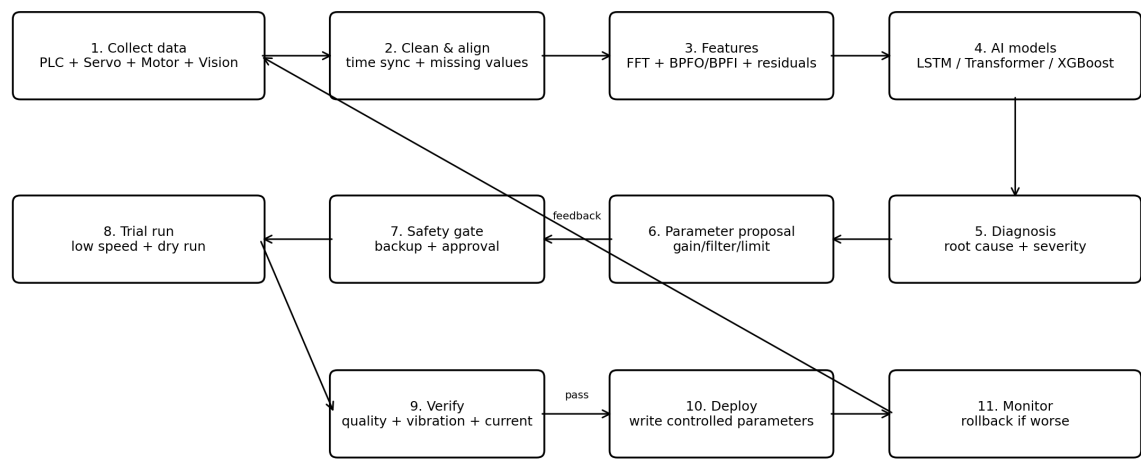


圖 2：AI 分析後回饋修改 Servo 參數流程

1. 系統規格與通訊串接

資料從 Level 0 的 Servo Drive、Motor、Encoder、振動/溫度感測器，經由 PLC/Motion Controller 與 HMI/SCADA，進入 AI Feature Store。AI Engine 不直接無條件寫入參數，而是產生建議、風險與驗證流程，由工程師審核後才進行 controlled write。

Layer	Device	Protocol/Interface	Main Data	Sampling/Cycle	AI Usage
Level 3	MES / Factory DB	SQL/REST/OPC UA	Lot, recipe, product, maintenance history	1s-1min	Context and labels
Level 2	SCADA / Historian	OPC UA / MQTT	Trend, alarms, event logs, OEE	100ms-1s	Feature store
Level 1	HMI	Ethernet / GOT / SCADA	Status, alarms, AI recommendation, approval button	100ms-1s	Human-in-the-loop approval
Level 1	PLC / Motion Controller	Ethernet, CC-Link IE TSN, EtherCAT gateway	Command, step, interlock, scan time	1-10ms	Motion context
Level 0	Servo Drive	SSCNET, EtherCAT, CC-Link IE, pulse/train, analog	position, speed, torque, current, temp, alarm	0.5-2ms	High-frequency AI signals
Level 0	Motor / Encoder / Sensor	Encoder, analog, IO-Link, accelerometer	encoder, vibration, bearing temp, limit sensor	0.5-10ms	Fault features

2. 資料分析後的性能提升與參數回饋

AI 判斷不應只停在 Alarm 分類，而要形成工程行動：先診斷 root cause，再選擇可調參數群，最後經過安全閘門、低速試跑、KPI 比較與 rollback 機制。

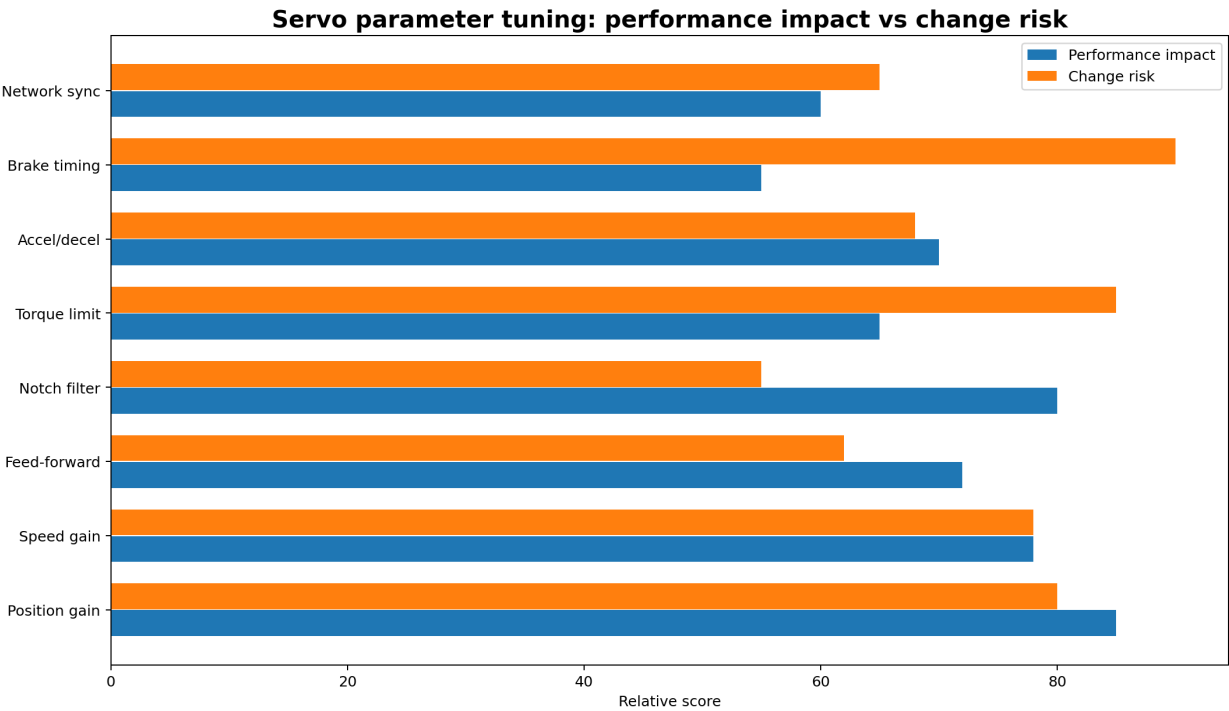


圖 3：參數調整的性能影響與風險比較

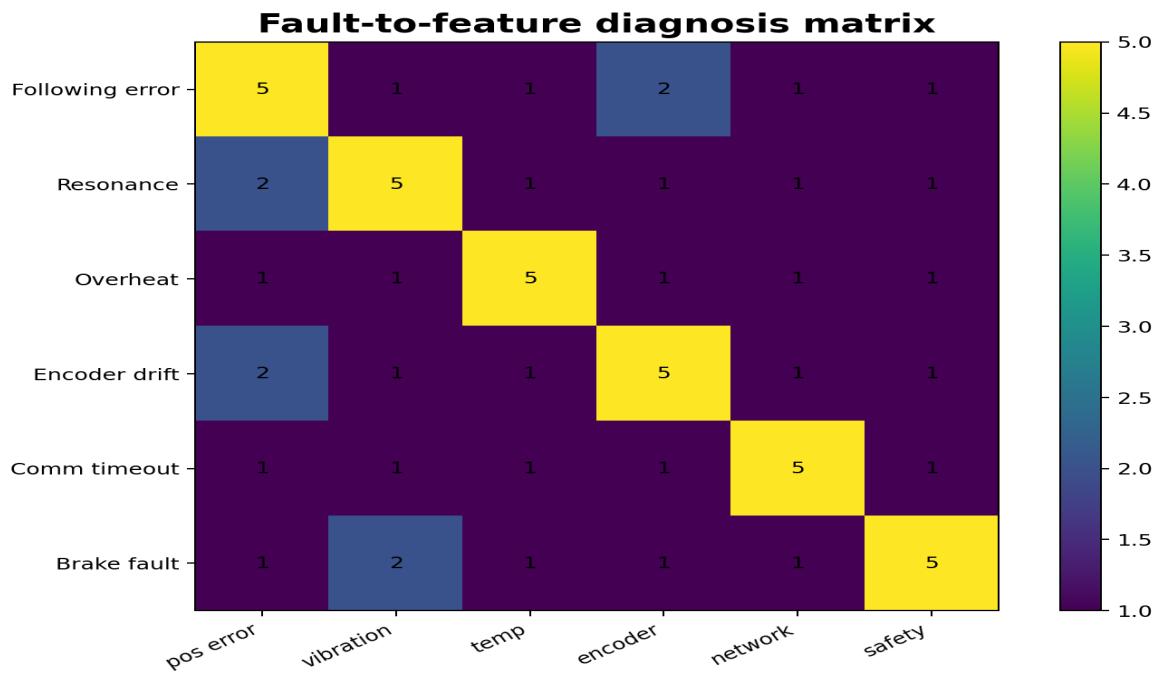


圖 4：Fault-to-Feature 診斷矩陣

3. 三菱 Servo 參數建議矩陣

以下採用 Mitsubishi-style 參數群概念。實際參數編號、單位、上下限與可寫入條件，必須依現場 MR-J4/MR-J5 實際型號與官方手冊確認。

Parameter Group	Purpose	AI Trigger	Direction of Change	Risk
Backup / restore	Create rollback point before tuning	Any AI write recommendation	Backup first, no tuning yet	Low
Auto tuning	Estimate inertia and basic gains	New equipment, changed load, high following error	Run tuning routine under safe mode	Medium
Response level	Improve settling time	Long settling, low vibration	Increase gradually	Medium
Position loop gain	Reduce position error	High following_error_abs_pulse	Increase small step	High
Speed loop gain	Reduce speed tracking error	speed_error_rpm high	Increase small step	High
Speed integral compensation	Improve low-speed tracking	low speed error / drift	Decrease carefully for faster response	Medium
Feed-forward	Reduce lag during acceleration	accel_error and position_error rise together	Increase gradually	Medium
Torque limit	Protect mechanism and reduce current trip	current_limit_pct high / jam risk	Lower limit for protection or raise if undersized after review	High
Current filter	Reduce current ripple/noise	current ripple and vibration noise	Increase filtering carefully	Medium
Notch filter	Suppress resonance frequency	resonance_amp high / FFT peak	Enable/tune notch around detected frequency	Medium
Vibration suppression	Reduce settling vibration	vibration_rms_g high after move	Enable suppression and tune frequency	Medium
Load inertia ratio	Improve model accuracy	AI inertia estimate changes	Update estimated ratio	Medium
In-position range	Avoid false complete or slow cycle	motion_complete mismatch	Tighten or relax based on process tolerance	Low
Following error limit	Early protection	position_error rising before alarm	Set warning/trip limits	High
Soft limit	Prevent collision	near limit / overtravel risk	Set according to stroke	High
Brake timing	Protect vertical Z axis	brake failure or slip detected	Adjust release/hold delay	High
Forced stop decel	Safety stop behavior	E-stop or STO events	Set safe decel profile	High

4. AI Servo 閉環規則

閉環規則會將診斷條件轉成參數建議，但每一條規則都必須經過權限控管與驗證流程。

Rule ID	Detection condition	AI diagnosis	Recommended action	Parameter candidates	Approval required
R01	position_error_pulse P95 > baseline x 3 and vibration low	Gain too low or load inertia changed	Run auto tuning and raise position/speed gain one step	Auto tuning, position loop gain, speed loop gain	Engineer + maintenance
R02	vibration_rms_g high and fft_1x/2x peak stable	Mechanical resonance or imbalance	Enable notch/vibration suppression, inspect mechanics	Notch filter, vibration suppression, speed gain	Engineer + process
R03	motor_temp_c slope high with current high	Thermal overload	Lower duty cycle/torque limit, inspect load	Torque limit, acceleration/deceleration time	Engineer + safety
R04	encoder_error_count drift, no vibration	Encoder drift/cable/noise	Inspect encoder cable, recalibrate home/absolute position	Encoder setting, home offset, filter	Engineer
R05	network_jitter_ms or packet_loss_pct high	Communication issue	Check EtherCAT/CC-Link network before servo tuning	Network sync, PLC scan, retry timeout	Controls engineer
R06	brake_release_status abnormal and Z axis speed deviation	Brake failure/slip	Stop line and inspect brake before parameter change	Brake timing only after mechanical check	Safety approval

5. 模型訓練與部署

資料可直接切窗給 LSTM、Transformer、Informer、PatchTST、AutoEncoder 使用；也可以以統計特徵、FFT 特徵與 Digital Twin residual 餵給 XGBoost、LightGBM、CatBoost。部署時建議先做 read-only advisory mode，再進入 supervised parameter recommendation，最後才進入 controlled write。

6. 安全聲明

伺服參數調整會影響機械安全、產品品質與人員安全。任何 AI 建議都必須先備份原參數、確認設備型號、低速空跑、確認 interlock 與 E-stop，再由授權工程師核准。

參考：Mitsubishi Electric MR-J5 User's Manual (Adjustment) 與 MR-J5-A User's Manual (Parameters)。